1 DIGITAL DEVICES

Digital devices are pieces of hardware that use computers or microcontrollers, and they are found everywhere in our digital world. They enhance and support how we live our lives every day. They can connect and work together to give us the data we need, when and where we need it.

Digital devices are always developing. This changes the way in which they are used by individuals, organisations and local, national and global societies.

LEARNING OBJECTIVES

- Be aware that mainframe computers are used for complex processing tasks and microprocessors are embedded in products such as washing machines
- Understand that laptop and desktop computers are types of personal computers. Some laptops are used as desktop replacements
- Know about types of mobile phones; smartphones and specialist phones and how they connect to the network (SIM)
- Know about tablet devices
- Be able to describe the purpose and use of other digital devices such as:
  - cameras and camcorders
  - games consoles
  - home entertainment systems
  - media players
- Know about navigation aids and how they are used
- Understand the terms 'multifunctional' (e.g. mobile phones that include a camera, have limited game playing functionality and GPS) and 'convergence' (e.g. functionality of smartphones and tablet devices becomes more similar) in the context of digital devices
- Understand features of digital devices: portability, performance, storage, user interface, connectivity, media support, energy consumption, expansion capability, security features
- Be able to discuss the features of identified digital devices
- Be able to identify digital devices and associated peripheral devices that meet particular needs, including accessibility
- Know about types of output peripheral such as monitor (screen size, resolution), printer (laser, inkjet, 3D), plotter, data projector, speaker, control device and when they would be used
- Know about types of input peripheral such as keyboard, mouse, tracker ball, joystick, graphics tablet, scanner, digital camera, webcam, microphone, touch screen, OMR reader, OCR reader, bar code scanner, biometric scanner, magnetic stripe reader, chip and pin, sensor and when they would be used
- Be able to differentiate between storage devices and the media used to store data
- Know the characteristics of hard disk drives (HDD), solid state drives (SSD), optical disk drives, flash memory drives
- Know that storage devices can be internal or external
- Know about types of storage media such as hard disks, optical disks (CD, DVD, Blu-ray), flash memory devices, magnetic tape
- Know that storage media can be recordable / write once (R) and rewritable (R/W)
- Understand the terms describing the capacity of storage such as bit, byte and multiples of these (kbytes, mbytes, gbytes, tbytes) (using 1KB = 1000 bytes)

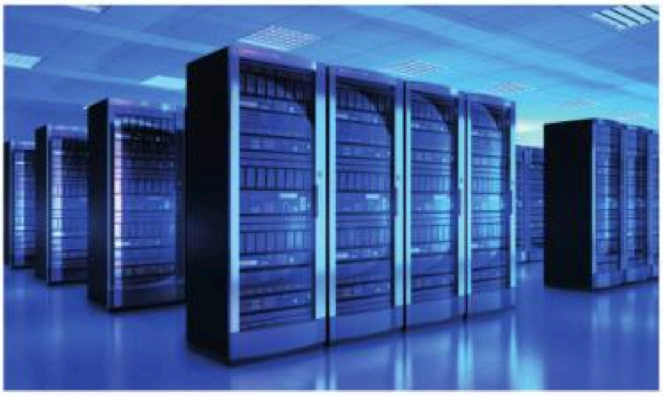
TYPES OF DIGITAL DEVICES

SUBJECT VOCABULARY

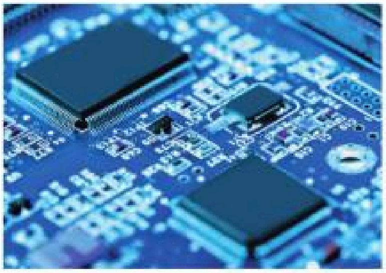
mainframe computer a large, powerful computer that can do a lot of complicated jobs quickly and can be used by a lot of people at the same time

microprocessor the device that controls what a computer does; it takes data as input, does something with it and provides output

There are many types of digital devices. They range from very powerful **mainframe computers**, used by large organisations for complex processing tasks such as statistical analysis and bulk data processing, to **microprocessors** used to control washing machines, televisions and other household appliances. Examples of the devices you will consider include personal computers, digital cameras and home entertainment systems.



▲ **Figure 1.1** A mainframe computer



▲ **Figure 1.2** A microprocessor

SKILLS CRITICAL THINKING
INTERPRETATION
COMMUNICATION

ACTIVITY

▼ DIGITAL DEVICES

1 Research the difference between a mainframe computer, a server and a supercomputer.

2 Create a list of the digital devices in your home that use microprocessors.

PERSONAL COMPUTERS

Personal computers (PCs) are common in homes and offices. They come in many different shapes and sizes, such as desktops and laptops.

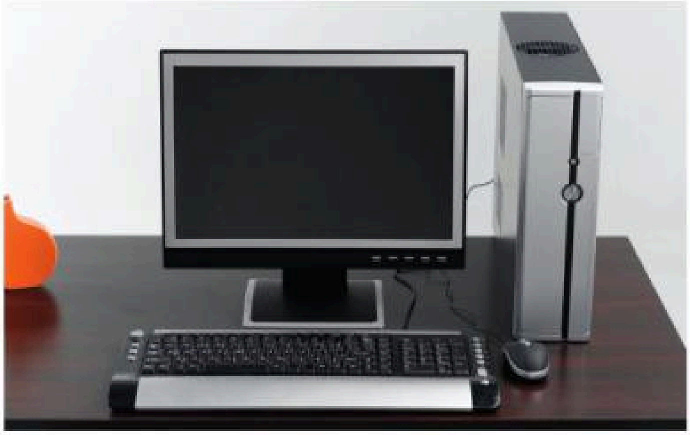
SUBJECT VOCABULARY

upgrade to make a computer better and able to do more things

peripheral device equipment that is connected to a computer and used with it

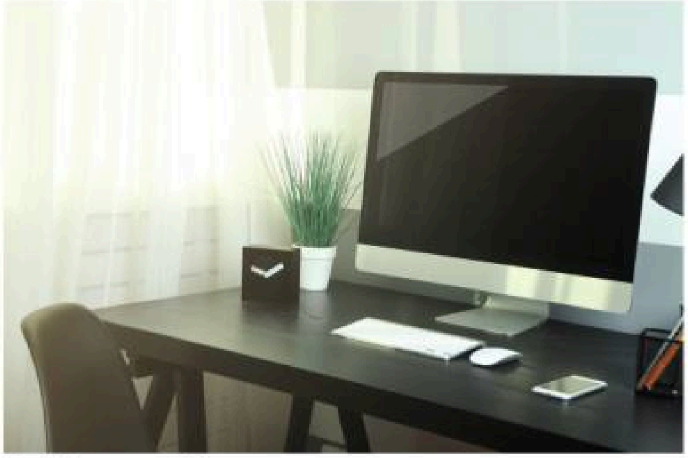
DESKTOPS

Desktops have more space for components than laptops and often provide users with the option to **upgrade** them or add additional components. A desktop computer usually needs to have **peripheral devices** connected to it, such as a monitor, a printer, a mouse and a keyboard. For more information about peripheral devices, see pages 19, 20 and 32.



▲ **Figure 1.3** A desktop computer

Some desktops are 'all-in-one'. This means that they combine the monitor with the PC hardware, as shown in Figure 1.4.



▲ **Figure 1.4** An all-in-one computer

LAPTOPS

Laptops include a keyboard, a screen, a track pad (see page 23) and a rechargeable battery. Having these features means that they can be taken away from the desk, which makes them an example of a **portable** device.

GENERAL VOCABULARY

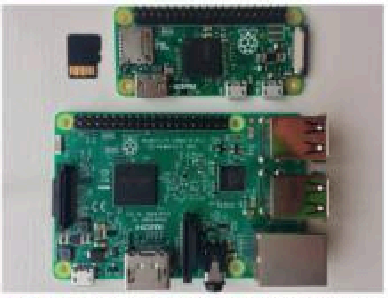
portable easy to carry or move

▲ **Figure 1.5** A laptop computer

Some laptops are called 'desktop replacements'. These tend to be larger than other laptops, as well as having a bigger and better-quality screen and higher performance levels.

SINGLE-BOARD COMPUTERS

Single-board computers (SBCs) are affordable computers used in education, **embedded computing** projects and **physical computing** projects. The Raspberry Pi Zero (see Figure 1.6) is an example of an SBC. The unit itself costs very little and uses a cheap microSD card as its storage.



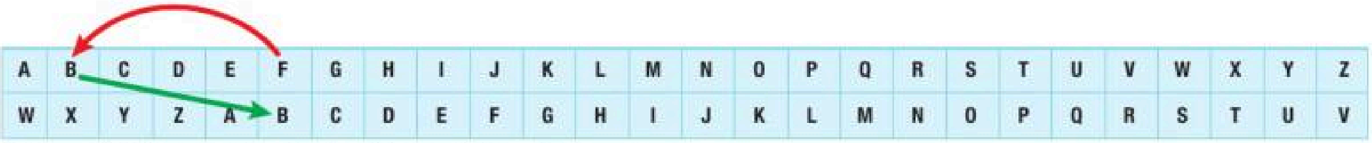
▲ **Figure 1.6** The Raspberry Pi Zero (top) and its microSD card (top left)

SUBJECT VOCABULARY

embedded computing computing hardware that is fixed into position and carries out a specialist task

physical computing interactive systems that can sense and respond to the world around them

MOBILE PHONES



A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V

▲ **Figure 5.18** Once you know the key, it is easy to decrypt the message

SKILLS PROBLEM SOLVING
INTELLECTUAL INTEREST AND CURIOSITY

ACTIVITY

▼ DECRYPTING A MESSAGE

Using Figure 5.18, decrypt this message: WHWJ PQNEJC. When you have decrypted it, you will find that it is a person's name. Research the importance of this person's work during the Second World War.

SKILLS PROBLEM SOLVING
INTELLECTUAL INTEREST AND CURIOSITY
ADAPTABILITY
CO-OPERATION
RESPONSIBILITY

ACTIVITY

▼ ENCRYPTING A MESSAGE

1. The following is a message encrypted with a Caesar cipher. The shift is 16.
   CEIJBO XQHCBUII
   Decipher the message to give the original text.
2. Use a new Caesar cipher key to create an encoded message for a classmate. Do not include spaces between words, because this makes it harder to decrypt the message. Give them the encrypted message and see whether they can decrypt it and tell you the key.

A Caesar cipher is quite easy to crack, but most modern encryption is much more secure. There are two types of encryption:

- symmetric key encryption
- public key encryption.

SYMMETRIC KEY ENCRYPTION

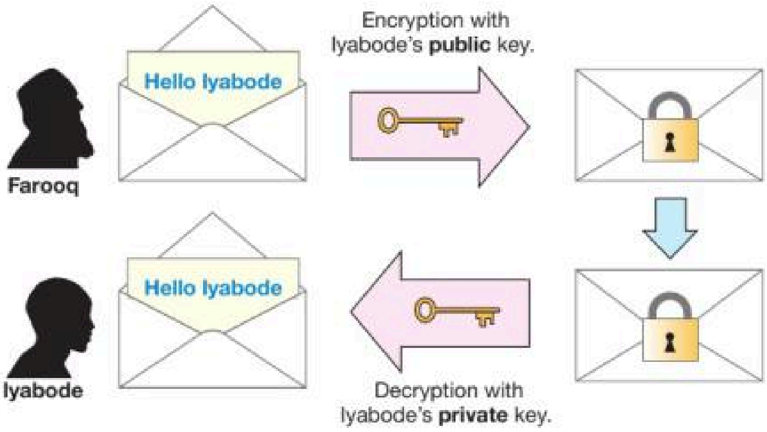
Symmetric key encryption uses the same key at both ends of the process, meaning that the same key is used to encrypt and decrypt the data.

PUBLIC KEY ENCRYPTION

Public key encryption uses two mathematically related keys called a key pair. One key is used to encrypt the data and a different key is used to decrypt it.

A computer shares a public key with other computers that want to send it encrypted data. This public key is mathematically related to a private key, which is not shared.

Figure 5.19 shows an example of public key encryption. If Farooq wants to send Iyabode an encrypted message, Farooq uses Iyabode's public key to encrypt the data. The data is then sent to Iyabode, who uses her private key to decrypt the message.



▲ Figure 5.19 Farooq first sends Iyabode a request to send a secure message, then Iyabode shares her public key with Farooq

HINT

A key is not the same as a password.

Although Farooq's message won't be secure, because the public key is published, anyone who receives the message will not be able to understand it without Iyabode's private key.

SUBJECT VOCABULARY

eavesdropper an unauthorised person or piece of software that intercepts data from a private communication
packet a unit of data packaged to travel across a network

WIRELESS ENCRYPTION PROTOCOL (WEP)

It is easier to intercept data in a wireless network than in a wired network. Wireless Encryption Protocol (WEP) is used to secure the wireless transfer of data. It is the least secure wireless data encryption method. This is because every device on the wireless network uses the same key for every transfer. This means that if an **eavesdropper** studies enough **packets**, they can identify the key, and this provides them with unlimited access to all data from every device on the wireless network.

DID YOU KNOW?

Experts tested the strength of WEP by examining 10,000 packets in less than a minute. Having done this, they then took less than 3 seconds to identify the WEP encryption key used to encrypt the packets.

WI-FI PROTECTED ACCESS (WPA)

Wi-Fi Protected Access (WPA) is a security protocol designed to provide better encryption than WEP. WPA generates a new key for each device on the wireless network. New keys are also provided for each packet of data that is sent.

VIRTUAL PRIVATE NETWORK (VPN)

A VPN provides access to a private LAN from a remote location. The connection to the LAN is created using the infrastructure of a public network like the internet. Data sent using a VPN is encrypted so that it is secure if it is intercepted. An individual might use a VPN to:

- access their employer's network when working from home
- access computers in a different geographical location, perhaps to avoid the local restrictions on access to web content (such as due to censorship or **geolocation rights management**)
- make secure payments
- prevent surveillance of and access to their web activity.

SUBJECT VOCABULARY

geolocation rights management the use of a user's location to block access to services online; for example, a television channel might only allow users in their own country to watch their shows online

DID YOU KNOW?
Mobile, desktop and browser apps are available to make it easy for users to change their network settings so that they can use VPNs.

FILE ACCESS RIGHTS

File access rights are also known as file permissions. They can be set for individual files, folders or drives, and they ensure that users are either allowed to read only or allowed to read and write to the file, folder or drive.

TRANSACTION LOGS

SUBJECT VOCABULARY

log a record of events
transaction the exchange of data

All network activity can be recorded in a **log** file. Although this does not directly secure network data, a **transaction** log can help to identify which computers and network devices have been accessed. This can allow administrators to identify any unusual activity that might be a threat to data security.

BACKUPS

A backup is a copy of one or more files. The backup or backups are usually stored on a different storage device to the original file.

For more information about backups, see *Unit 1 Digital devices* (page 42) and *Unit 3 Operating online* (pages 104–105).

CHAPTER QUESTIONS

SKILLS PROBLEM SOLVING

1 Which **one** of these connects a LAN to a WAN? (1)

A Switch
B Gateway
C Modem
D Server

SKILLS DECISION MAKING

2 Explain why public key encryption is more secure than symmetric key encryption. (2)

SKILLS PROBLEM SOLVING

3 State which security device controls the traffic entering a network. (1)

SKILLS PROBLEM SOLVING

4 State **two** methods by which devices are identified to each other on a network. (2)

SKILLS REASONING

5 Explain why IPv6 was used to replace IPv4. (2)

SKILLS PROBLEM SOLVING

6 State **two** uses for a VPN. (2)

SKILLS REASONING, PROBLEM SOLVING

7 State the reason why a booster is used in a network. (1)

SKILLS INTERPRETATION

8 Describe how encryption secures data. (3)

SKILLS PROBLEM SOLVING

9 List **three** methods of securing data on networks. (3)

SKILLS DECISION MAKING

10 State **three** benefits to users of using a local area network. (3)

SKILLS ADAPTIVE LEARNING, EXECUTIVE FUNCTION, ADAPTABILITY, COMMUNICATION

11 Draw and label a network diagram for a home that includes:

- two smartphones
- one tablet device
- two desktop PCs
- internet access. (8)